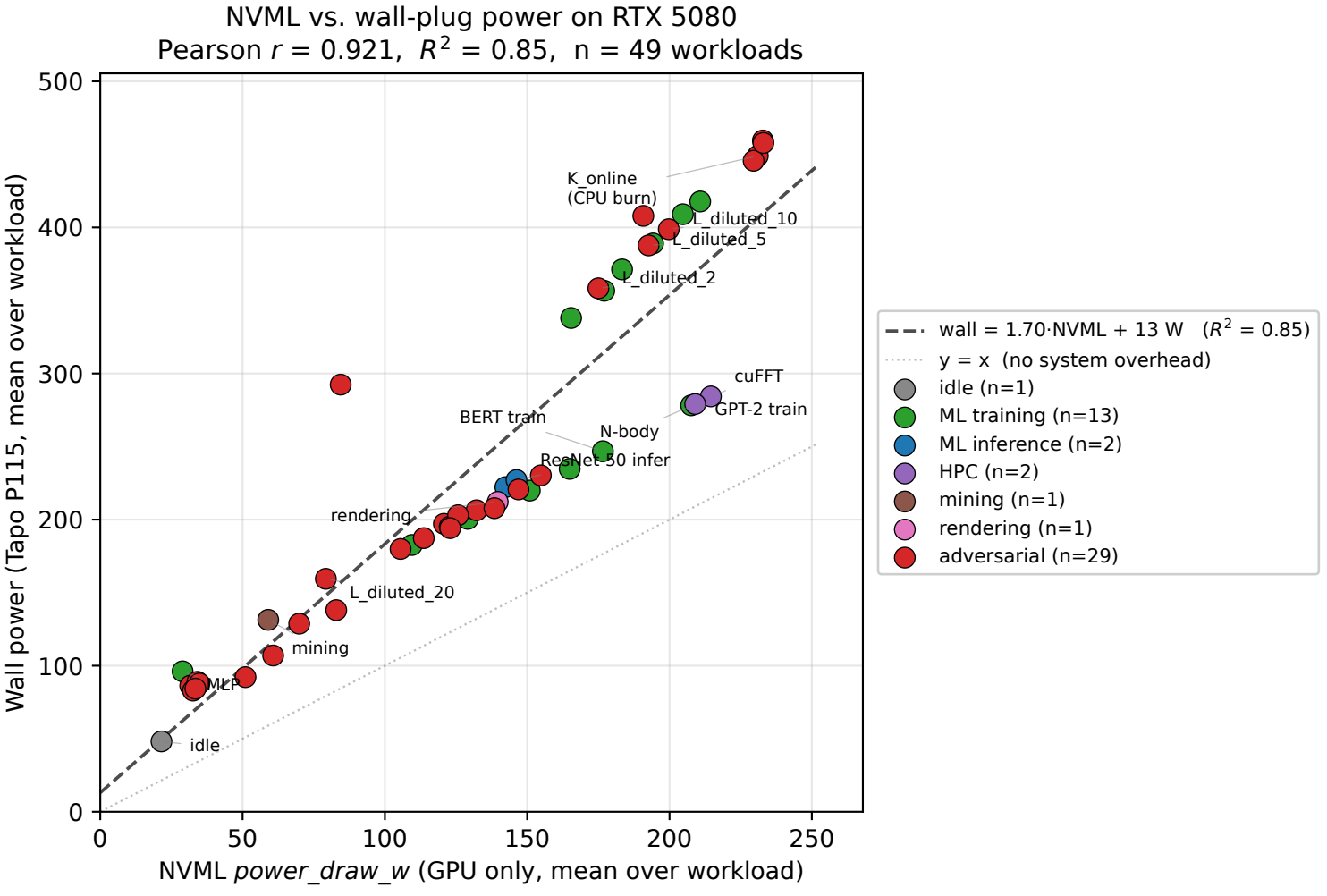
NVML vs. wall-plug power on RTX 5080
Pearson r = 0.921, R² = 0.85, n = 49 workloads
Wall power (Tapo P115, mean over workload)
NVML power_draw_w (GPU only, mean over workload)
0
100
200
300
400
500
0
50
100
150
200
250
K_online
(CPU burn)
L_diluted_10
L_diluted_5
L_diluted_2
cuFFT
GPT-2 train
BERT train
N-body
ResNet-50 infer
rendering
L_diluted_20
mining
MLP
idle
wall = 1.70·NVML + 13 W (R² = 0.85)
y = x (no system overhead)
idle (n=1)
ML training (n=13)
ML inference (n=2)
HPC (n=2)
mining (n=1)
rendering (n=1)
adversarial (n=29)